A

B

C

Signal-to-Noise Ratio ($\rho^2/\sigma^2$)

Spike Strength ($\gamma$)

Generalization Error Difference ($\Delta\tilde{G}$)

no feature learning advantage

feature learning advantage

Generalization Error

A

B

C

Ridge ($\lambda$)

$\rho^2/\sigma^2 = \dfrac{(1+\gamma\psi)^3}{\gamma(1+\gamma)(1-\psi)^3}$ (upper bound)

$\rho^2/\sigma^2 = \dfrac{(1+\gamma\psi)^2}{\gamma(1+\gamma)(1-\psi)^2}$ (lower bound)

Sufficient SNR Window

Feature Learning Advantage Threshold ($\Delta\tilde{G}=0$)

baseline kernel

one-step kernel